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import matplotlib.pyplot as plt

# 1. JSW STEEL -- REAL EMISSIONS AND PRODUCTION (FY2025-26)

# Source: JSW Steel Limited official BRSR XBRL filing, NSE
(nseindia.com),
# FY2025-26. Extracted and verified directly from the filed XML in
this session.
JSW_SCOPE1_TC02E = 60_258_642

# Source: JSW Steel FY26 quarterly/annual production releases
(jswsteel.in).
# India operations production, FY2025-26.
JSW_INDIA_PRODUCTION_T = 29_250_000

jsw_intensity = JSW_SCOPE1_TC02E / JSW_INDIA_PRODUCTION_T

# Cross-check (SOFTENED, not diagnostic): JSW publicly discloses ~2.4
tCO2/tcs
# for combined Scope 1+2 at its integrated plants. Ohio is <3% of
JSW's
# consolidated volume (0.83 of 30.14 Mt) -- even at a low EAF
intensity of
# ~0.5 tCO2/t for Ohio, the consolidated blend (2.013) sits only ~2%
below
# the India-only figure (2.060), well inside the uncertainty of the
assumed
# ~0.3-0.4 Scope 2 addition. This is CONSISTENT WITH an India-only
numerator
# but the volumes are too close for this check to DISCRIMINATE between
the
# two hypotheses. Stated as weak supporting evidence only.
print(f"JSW Scope 1 intensity (India ops, entity-level BRSR, inferred
boundary): {jsw_intensity:.3f} tCO2e/t")
print("Cross-check vs disclosed ~2.4 tCO2/tcs (Scope1+2): consistent
with, not diagnostic of, India-only boundary")

JSW Scope 1 intensity (India ops, entity-level BRSR, inferred
boundary): 2.060 tCO2e/t
Cross-check vs disclosed ~2.4 tCO2/tcs (Scope1+2): consistent with,
not diagnostic of, India-only boundary

# 2. EU BENCHMARK VALUE (used in free-allocation / SIFA calculation)

# Source: EUROMETAL report on leaked/finalized Implementing Regulation
(EU)
# 2025/2621 document (Dec 2025), corroborated by two secondary CBAM
# compliance-guidance sites citing the same IR. BF-BOF benchmark =
1.370

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# tCO2e/t of crude steel.

# TWO UNRESOLVED RISKS, both material to the headline finding:
# (1) UNVERIFIED whether the definitive phase applies a DIFFERENT
# (lower)
#     benchmark specifically when actual emissions data is submitted,
#     vs.
#     when EU default values are used. This model applies 1.370 to
#     BOTH
#     branches below. If the regulation is asymmetric, this would
#     compress
#     or reverse the actual-vs-default cost gap that is this project's
#     central finding.
# (2) UNVERIFIED whether 1.370 is the benchmark for CRUDE STEEL / hot
#     metal,
#     or specifically for COLD-ROLLED FLAT STEEL (JSW's actual export
#     product and CN heading). CBAM benchmarks are set per CN code in
#     the
#     Annex; a mismatch here changes the result directly. Sensitivity:
#     each 0.01 tCO2e/t of benchmark moves cost by ~€0.73/t -- an
#     earlier
#     correction from 1.328 to 1.370 alone moved the headline number
#     by 5.3%.
STEEL_BENCHMARK = 1.370 # UNVERIFIED for CN-code match; UNVERIFIED
# for actual/default symmetry

# 3. EU DEFAULT VALUE (used when NO actual data is submitted)
# ILLUSTRATIVE, LIKELY CONSERVATIVE in the direction that matters:
# China's
# CONFIRMED steel-slab default under IR 2025/2621 is 3.167 tCO2e/t.
# The
# definitive-phase rule sets defaults for countries lacking their own
# reliable data at the HIGHEST emission intensity observed among
# countries
# WITH reliable data -- so if no India-specific value exists, India's
# default is likely AT OR ABOVE 3.167, not below the 2.5-based figure
# here.
# This estimate is therefore a FLOOR on the true saving, not a
# ceiling.
BASE_DIRECT_HIGH_UNVERIFIED = 2.5
MARKUP_2026 = 0.10 # Source: per IR 2025/2621 -- 10% (2026), 20%
# (2027), 30% (2028+)
DEFAULT_HIGH_2026_UNVERIFIED = BASE_DIRECT_HIGH_UNVERIFIED * (1 +
MARKUP_2026)

# China-anchored sensitivity -- more defensible upper bound given the
# "highest observed intensity" rule
CHINA_DEFAULT_CONFIRMED = 3.167 # Source: IR 2025/2621, confirmed
# via EUROMETAL/CBAM Guide

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CHINA_DEFAULT_WITH_MARKUP = CHINA_DEFAULT_CONFIRMED * (1 +
MARKUP_2026)

# 4. FREE ALLOCATION SHARE / SIFA

# Source: Article 10a(1a), Directive 2003/87/EC -- CBAM factor for
2026
# is 2.5% (i.e. 97.5% of free EU ETS allocation still applies).
CBAM_FACTOR_2026 = 0.025
FREE_ALLOC_SHARE_2026 = 1 - CBAM_FACTOR_2026    # 0.975

# UNVERIFIED -- Cross-Sectoral Correction Factor asserted at 1.0 (i.e.
not
# reducing free allocation further); not confirmed whether CSCF
applies to
# the CBAM free-allocation adjustment at all, or only to EU ETS
internally.
CSCF = 1.

# SIFA = the free-allocation deduction subtracted from gross embedded
# emissions before certificates are owed. This construction
(subtracting
# SIFA from emissions, THEN multiplying by price) is the correct
mechanism
# -- NOT the common shortcut of "cost = emissions x price x 2.5%",
which is
# only accurate for a producer sitting exactly at benchmark
efficiency.
sifa_2026 = FREE_ALLOC_SHARE_2026 * CSCF * STEEL_BENCHMARK
print(f"\nSIFA (free allocation adjustment), 2026: {sifa_2026:.5f}
tCO2e/t")

def certs_per_tonne(SE_direct, sifa_val):
    """Certificates required per tonne = embedded emissions minus free
allocation, floored at zero (can't owe negative certificates)."""
    return max(SE_direct - sifa_val, 0)

SIFA (free allocation adjustment), 2026: 1.33575 tCO2e/t

# 5. OFFICIAL CBAM CERTIFICATE PRICE

# Source: European Commission DG TAXUD CBAM certificate price page --
# weighted average EU ETS auction clearing price, published quarterly.
# Applies specifically to goods imported during the stated quarter.
EUA_PRICE_Q1_2026 = 75.36    # €/tCO2e, Q1 2026 (Jan-Mar)

# 6. HEADLINE FINDING: ACTUAL VS DEFAULT

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certs_actual = certs_per_tonne(jsw_intensity, sifa_2026)
certs_default = certs_per_tonne(DEFAULT_HIGH_2026_UNVERIFIED,
sifa_2026)

cost_actual = certs_actual * EUA_PRICE_Q1_2026
cost_default = certs_default * EUA_PRICE_Q1_2026
saving_pct = (1 - cost_actual / cost_default) * 100

print(f"\nCBAM cost, ACTUAL verified data:                €
{cost_actual:.2f}/t")
print(f"CBAM cost, EU DEFAULT (unverified upper band):  €
{cost_default:.2f}/t")
print(f"Saving from using verified data vs default:
{saving_pct:.1f}%")
print("CAVEAT: this saving assumes the SAME benchmark applies to both
branches -- unconfirmed (see Section 2).")

CBAM cost, ACTUAL verified data:                €54.59/t
CBAM cost, EU DEFAULT (unverified upper band):  €106.58/t
Saving from using verified data vs default:      48.8%
CAVEAT: this saving assumes the SAME benchmark applies to both
branches -- unconfirmed (see Section 2).

# China-anchored sensitivity
certs_china = certs_per_tonne(CHINA_DEFAULT_WITH_MARKUP, sifa_2026)
cost_china = certs_china * EUA_PRICE_Q1_2026
saving_china_pct = (1 - cost_actual / cost_china) * 100

print(f"\nSENSITIVITY -- if India's default tracks the confirmed China
anchor (3.167, +10% markup):")
print(f"  CBAM cost at China-anchored default: €{cost_china:.2f}/t")
print(f"  Saving from actual data: {saving_china_pct:.1f}%")
print(f"  The {saving_pct:.1f}% figure above is a FLOOR, not a ceiling
-- if India's true default")
print(f"  is closer to the China anchor, the real saving from verified
data is materially higher.")

SENSITIVITY -- if India's default tracks the confirmed China anchor
(3.167, +10% markup):
  CBAM cost at China-anchored default: €161.87/t
  Saving from actual data: 66.3%
  The 48.8% figure above is a FLOOR, not a ceiling -- if India's true
default
  is closer to the China anchor, the real saving from verified data is
materially higher.

# 7. CCTS -- CORRECT TIMING FRAMING

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# Source: iron & steel sectoral GEI (emission intensity) notification
under
# India's Carbon Credit Trading Scheme was issued June 2026, mandating
# compliance from FY2026-27 onward. The MODELLED LIABILITY here is for
# calendar-2026 imports:
#   - Jan-Mar 2026 imports fall in FY2025-26 -> CCTS does NOT yet
apply
#   -> zero deduction is the CORRECT current base case for this
period.
#   - Apr-Dec 2026 imports fall in FY2026-27 -> CCTS's first
compliance
#   year applies (subject to the caveat below).
print(f"\nCCTS timing: Jan-Mar 2026 imports (FY25-26) -- zero
deduction, correct base case.")
print(f"Apr-Dec 2026 imports (FY26-27) -- CCTS's first compliance year
applies, IF EU-recognised.")

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CCTS timing: Jan-Mar 2026 imports (FY25-26) -- zero deduction, correct
base case.
Apr-Dec 2026 imports (FY26-27) -- CCTS's first compliance year
applies, IF EU-recognised.

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# STRONGER CAVEAT than timing: Article 9 of Regulation (EU) 2023/956
requires
# the EU to formally recognise a third-country carbon pricing scheme
as
# equivalent before any deduction is permitted. CCTS has NOT been
confirmed
# as EU-recognised as of this analysis. The deduction below may not
apply
# in practice for ANY part of 2026, regardless of CCTS's domestic
timing.
print("CAVEAT: EU recognition of CCTS as a deductible scheme under
Article 9 is unconfirmed --")
print("the deduction shown below may not be legally available for any
2026 imports.")

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def cbam_cost_with_ccts(SE_direct, sifa_val, eua_price,
origin_price_inr_per_tco2e, gap_share):
    """Models a domestic carbon price deduction, IF the EU recognises
CCTS.
    Deduction is capped at what CBAM actually bills (cannot deduct
more
    than gross certificate cost) -- avoids double-counting against
SIFA."""
    gross_certs = certs_per_tonne(SE_direct, sifa_val)
    gross_cost = gross_certs * eua_price
    origin_price_eur = origin_price_inr_per_tco2e / 95    # EUR_INR =
95, STATED RATE, Aug 2026 -- not live/dynamic

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    deduction = min(origin_price_eur * SE_direct * gap_share,
origin_price_eur * gross_certs)
    return max(gross_cost - deduction, 0)

```

CAVEAT: EU recognition of CCTS as a deductible scheme under Article 9 is unconfirmed --
the deduction shown below may not be legally available for any 2026 imports.

```

# Source: CCTS iron & steel GEI targets require intensity reductions
ranging
# 2.1%-9.3% across obligated entities, median ~5.5% (per the June 2026
# notification). Used here as the "gap share" -- the share of an
entity's
# emissions intensity that must close to meet target, NOT the share of
# emissions actually priced under CBAM.
# ₹500/tCO2e is ILLUSTRATIVE ONLY -- no established Indian carbon
credit
# market price exists yet; CCTS trading infrastructure is still
standing up.
fy27_scenarios = [
    ("Apr-Dec 2026, median target (5.5% gap), ₹500/t illustrative",
500, 0.055),
    ("Apr-Dec 2026, steepest target (9.3% gap), ₹500/t illustrative",
500, 0.093),
]
print("\nIF CCTS were EU-recognised for Apr-Dec 2026 imports
(hypothetical):")
for label, price, gap in fy27_scenarios:
    cost = cbam_cost_with_ccts(jsw_intensity, sifa_2026,
EUA_PRICE_Q1_2026, price, gap)
    print(f"    {label}: €{cost:.2f}/t")

```

IF CCTS were EU-recognised for Apr-Dec 2026 imports (hypothetical):
 Apr-Dec 2026, median target (5.5% gap), ₹500/t illustrative:
 €53.99/t
 Apr-Dec 2026, steepest target (9.3% gap), ₹500/t illustrative:
 €53.58/t

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# 8. CHART -- ACTUAL VS DEFAULT
fig, ax = plt.subplots(figsize=(7,6))
labels = ["Actual verified\data", "EU default\n(illustrative – likely
conservative)"]
values = [cost_actual, cost_default]
colors = ["#2E7D32", "#B71C1C"]
ax.bar(labels, values, color=colors)
ax.set_ylabel("CBAM cost (€ per tonne)")
ax.set_title("JSW Steel: CBAM Cost – Actual vs EU Default (2026)")
for i, v in enumerate(values):

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ax.text(i, v + 1, f"€{v:.0f}", ha="center", fontweight="bold")

footnote = (
    f"Benchmark: {STEEL_BENCHMARK} tCO2e/t BF-B0F, per IR 2025/2621 --  

    UNVERIFIED for CN-code\n"
    f"match and for symmetry across actual/default branches. SIFA:  

    {sifa_2026:.5f} tCO2e/t.\n"
    f"Price: EUR{EUA_PRICE_Q1_2026}/tCO2e, Q1 2026 (DG TAXUD).  

    Emissions: JSW BRSR FY2025-26,\n"
    f"Scope 1, India ops (inferred boundary, not confirmed in filing).  

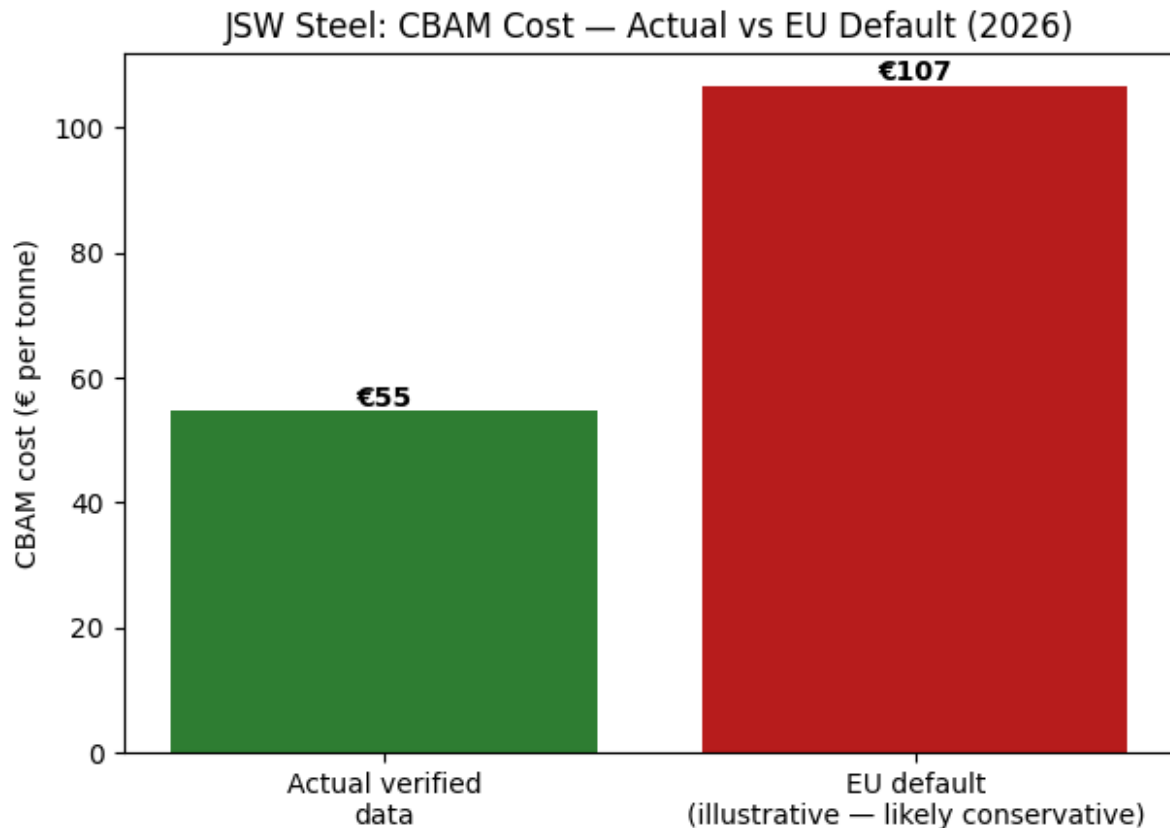
    CCTS: not applied --\n"
    f"EU recognition unconfirmed. Default value: illustrative  

    estimate, LIKELY CONSERVATIVE --\n"
    f"confirmed China default (3.167) exceeds this ceiling; India's  

    true default (if lacking own\n"
    f"data) may track China's per the 'highest observed intensity'  

    rule under IR 2025/2621."
)
fig.subplots_adjust(bottom=0.28)
fig.text(0.5, 0.06, footnote, ha="center", fontsize=7.5)
plt.savefig("cbam_actual_vs_default.png", dpi=150,
bbox_inches="tight")
plt.show()

```



Benchmark: 1.37 tCO₂e/t BF-BOF, per IR 2025/2621 -- UNVERIFIED for CN-code match and for symmetry across actual/default branches. SIFA: 1.33575 tCO₂e/t. Price: EUR75.36/tCO₂e, Q1 2026 (DG TAXUD). Emissions: JSW BRSR FY2025-26, Scope 1, India ops (inferred boundary, not confirmed in filing). CCTS: not applied -- EU recognition unconfirmed. Default value: illustrative estimate, LIKELY CONSERVATIVE -- confirmed China default (3.167) exceeds this ceiling; India's true default (if lacking own data) may track China's per the 'highest observed intensity' rule under IR 2025/2621.

9. LANDED COST STACK -- CBAM + ANTI-DUMPING DUTY (the headline number)

Source: European Commission General Disclosure document, 6 Aug 2026, # reported by S&P Global Platts / EUROMETAL ("EU announces definitive AD # duties on cold-rolled steel imports from five countries," eurometal.net, # 7 Aug 2026). Names JSW Steel Limited, JSW Steel Coated Products Limited, # and Tata Steel India specifically at 9.5%, following a 10-month # anti-dumping investigation initiated Sept 2025 on a EUROFER complaint.

AD_DUTY_RATE = 0.095

Source: same EUROMETAL article -- Platts price assessments, 6 Aug 2026.

```

CRC_CIF_PRICE = 655          # €/t, imported CRC, CIF Antwerp
CRC_DOMESTIC_PRICE = 825     # €/t, domestic CRC, ex-works Ruhr
domestic_import_spread = CRC_DOMESTIC_PRICE - CRC_CIF_PRICE

ad_duty_cost = CRC_CIF_PRICE * AD_DUTY_RATE

print(f"\n{'='*60}")
print(f"HEADLINE FINDING -- Landed cost stack per tonne CRC, 2026:")
print(f"\n{'='*60}")
print(f"  Anti-dumping duty (9.5%, EC 6 Aug 2026):  €
{ad_duty_cost:.2f}")
print(f"  CBAM (actual verified data, Q1 2026):      €
{cost_actual:.2f}")
print(f"  TOTAL:                                           €{ad_duty_cost +
cost_actual:.2f}")
print(f"  Domestic-import price spread (Platts):      €
{domestic_import_spread}")
print(f"  Remaining competitiveness headroom:          €
{domestic_import_spread - (ad_duty_cost + cost_actual):.2f}")
print(f"\nNote: this headroom applies at the 2026 CBAM factor (2.5%).
The factor rises to")
print(f"48.5% by 2030 and 100% by 2034 as EU free allocation phases
out -- the CBAM line")
print(f"alone will grow substantially even if the AD duty and steel
prices hold constant.")

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HEADLINE FINDING -- Landed cost stack per tonne CRC, 2026:
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Anti-dumping duty (9.5%, EC 6 Aug 2026):	€62.23
CBAM (actual verified data, Q1 2026):	€54.59
TOTAL:	€116.81
Domestic-import price spread (Platts):	€170
Remaining competitiveness headroom:	€53.19

Note: this headroom applies at the 2026 CBAM factor (2.5%). The factor rises to 48.5% by 2030 and 100% by 2034 as EU free allocation phases out -- the CBAM line alone will grow substantially even if the AD duty and steel prices hold constant.